

Data Management (A.A. 2025/26) – exam B of 04/06/2026

Problem 1 Let x and y be two elements of the database such that $rts(x) = rts(y) = wts(x) = wts(y) = wts-c(x) = wts-c(y) = 0$ and $cb(x) = cb(y) = true$. Suppose that the system clock is 0, and that the system uses the clock values as the timestamps to be assigned to the various transactions (at their first action). Illustrate the actions of the timestamp-based scheduler when receiving the following complete schedule:

$$r_1(x) \ r_2(y) \ c_2 \ w_1(y) \ c_1 \ r_3(x) \ w_3(y) \ c_3$$

and tell whether the resulting sequence of actions (obtained by ignoring aborted transactions) is a rigorous schedule.

Problem 2 Consider the following schedule S:

$$r_1(y) \ r_2(x) \ w_1(y) \ r_1(x) \ w_2(y) \ w_3(y) \ r_4(x) \ w_4(x) \ r_3(x)$$

- 2.1 Determine whether S is conflict-serializable or not, explaining your answer in detail.
- 2.2 Determine whether S is a 2PL schedule with exclusive locks. If it is not, determine whether adding shared locks would make it a 2PL schedule. Justify your answers in detail.
- 2.3 Determine whether S a strict schedule. Justify your answers in detail.

Problem 3 Let $R(A, B)$ and $S(A, B)$ be two bags with 6.000 and 12.000 pages, respectively. Assume that exactly 3 buffer frames are available. For each of the following situations:

- 3.1 R and S are not sorted;
- 3.2 R and S are sorted on all attributes;
- 3.3 R and S are sorted on attribute A ;
- 3.4 R and S are sorted on attribute A , and no two tuples of R having the same value of attribute A are stored in different pages. The same property holds for relation S .

illustrate which algorithm you would choose for computing the set difference $R - S$, and tell which is the cost of the algorithm in terms of number of page accesses.

Problem 4 Let $R(A, B, C, D)$ be a relation stored in a heap with 36.000 pages. Assume that all attribute values have the same size and that 140 buffer frames are available. Consider the following query:

select A, sum(D) from R group by A

Describe the most efficient algorithm that can be used to compute the query. Justify your choice and compute its cost in terms of number of page accesses.

Problem 5 (only for students who opted for **option 1**, i.e., who do **not** do the project)

- 5.1 Write an RDFS dataset G that expresses the following information. Use the RDF/RDFS vocabulary where appropriate; otherwise, use any URI you prefer (assuming an empty prefix):
 - Some reviewer has announced that the Statue of David was insured by a private collector for 500 million euros.
 - The Statue of David was sculpted by Michelangelo.
 - Michelangelo is also the name of a computer virus.
 - Everything that is crafted is an artifact.
 - Everyone who crafts is an artisan.
 - For every A, B , if A is sculpted by B , then A is crafted by B .
 - Artisan and reviewer are creative professions.
 - Every creative profession is a profession.
- 5.2 Write the SPARQL query Q that retrieves all artifacts.
- 5.3 Provide the answers to Q when it is (a) evaluated over the dataset G interpreted as a graph database, and (b) computed over G interpreted as an RDFS Knowledge Graph. Justify your answers.